

Towards Lightweight Defense against Gradient-Based Adversarial Attacks in Internet of Medical Things (IoMT) Systems

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Abstract—Medical imaging workflows in the Internet of Medical Things (IoMT) transmit sensitive images over networks that may be partially trusted. Adversaries can inject imperceptible ℓ_∞ -bounded perturbations during transmission, leading downstream models to misdiagnose. This paper introduces *Weighted Adversarial Example Training (WAET)*, a training-time defense that blends clean and adversarial objectives and emphasizes harder adversarial samples, while adding no computational overhead during inference. Experiments on the BloodMNIST datasets with DeiT-Tiny, MobileNet-V2, and ResNet-34 show that our WAET consistently improves robustness against the Fast Gradient Sign Method (FGSM), Basic Iterative Method (BIM), and Projected Gradient Descent (PGD) attack, across multiple perturbation budgets, while largely preserving clean accuracy; many of these gains remain statistically significant across repeated runs.

Index Terms—IoMT security, adversarial examples, adversarial training, robust medical imaging, gradient-based attacks

I. INTRODUCTION

With the support of the Internet of Medical Things (IoMT), medical devices, edge gateways, and Artificial Intelligence (AI) services can form an end-to-end diagnostic pipeline. This integrated setup facilitates automated medical image analysis and decision support in remote or resource-constrained environments, reducing turnaround time. However, moving inference to the edge broadens the attack surface for sensitive data: attackers could impersonate devices and launch man-in-the-middle attacks, while weak encryption or authentication leaves edge gateways as vulnerable interception points [1].

Adversarial examples introduce further risks: small, carefully chosen perturbations can mislead neural networks and potentially lead to harmful clinical decisions [2]. Many existing defenses are impractical in IoMT settings: input purification techniques can be brittle or induce latency, and strong adversarial training is computationally expensive and often overfits to specific attacks [3]. This paper proposes *Weighted Adversarial Example Training (WAET)*, a defence applied only during training to meet these constraints. WAET combines the losses of clean and adversarial examples, giving extra weight to harder adversarial samples so that the model learns more

robust decision boundaries. Importantly, this method changes only the offline training phase; the inference phase uses the same forward pass as a standard classifier and adds no extra runtime cost. WAET employs a three-step projected gradient descent (PGD) attack during training to reduce computational burden while keeping the deployed model lightweight. We select three PGD steps as a pragmatic compromise: additional iterations provide only marginal robustness gains while significantly increasing training time.

Our contributions are summarized as follows: (i) We formalize a realistic transmission-layer adversary for IoMT medical imaging and highlight the resource constraints that influence defence design; (ii) We introduce WAET, a light-weight adversarial training method that emphasizes harder adversarial samples and employs a three-step PGD during training, yet incurs no additional computation during inference; (iii) We evaluate WAET on the BloodMNIST datasets using DeiT-Tiny, MobileNet-V2, and ResNet-34. Across FGSM, BIM, and PGD attacks and multiple perturbation budgets, WAET improves robust accuracy without degrading performance on clean data.

II. SYSTEM AND THREAT MODEL

System Model: In the system, medical sensing devices (e.g., dermatoscopes and digital microscopes) acquire raw patient data and perform minimal preprocessing. The data are then forwarded to an *edge gateway*, which aggregates traffic, applies lightweight filtering, and relays the images to the cloud. Finally, a cloud (or edge) server hosts a deep neural network that performs inference and returns diagnostic predictions. This architecture is representative of practical deployments, in which computationally intensive inference is executed on servers while energy-constrained devices focus on acquisition and communication.

In our setting, training and inference are decoupled. WAET is used during offline training to harden the model. Then, the resulting parameters are deployed without changing the online inference pipeline. Denote $x \in \mathbb{R}^{H \times W \times C}$ as an input medical image and $y \in \{1, \dots, K\}$ as its label. The deployed model f_θ maps an image to class probabilities $f_\theta(x)$, and the system outputs $\hat{y} = \arg \max_k f_\theta(x)_k$.

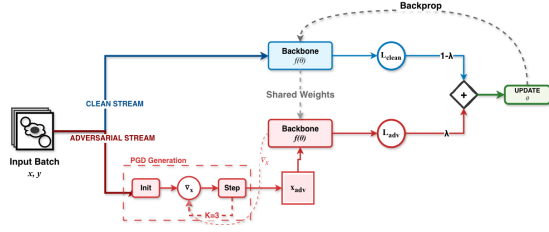


Fig. 1: WAET workflow

Threat Model: We consider a transmission-layer adversary with man-in-the-middle capabilities. The attacker can intercept images in transit (e.g., between the device and gateway, or between the gateway and cloud), and add small perturbations. Formally, the adversary produces a perturbed image $\tilde{x} = x + \delta$, $\|\delta\|_\infty \leq \epsilon$, where ϵ bounds the maximum per-pixel change. The adversary’s goal is to induce misclassification; however, the attacker has no access to model parameters, cannot modify the deployed model, and does not obtain privileged access to the cloud service. Such attacks can be mounted in a black-box fashion by crafting perturbations on a surrogate model and exploiting transferability. To assess robustness under this threat, we evaluate models using standard first-order gradient-based attacks (FGSM, BIM, and PGD). Our objective is to improve robustness at training time without adding inference-time overhead on resource-constrained gateways.

III. OUR APPROACH

Problem Formulation: Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote a training set of N medical images $x_i \in \mathbb{R}^{H \times W \times C}$ with labels $y_i \in \{1, \dots, K\}$. Our goal is to learn a classifier $f_\theta: \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^K$, parameterised by θ , that achieves high accuracy on clean inputs while remaining resilient to small adversarial perturbations.

Specifically, we assume an attacker can perturb each image by adding a noise vector δ such that $\|\delta\|_\infty \leq \epsilon$, where $\epsilon > 0$ denotes the adversary’s budget, and the perturbed input is $\tilde{x}_i = x_i + \delta$. Robustness requires that the classifier’s prediction remain correct under such perturbations, i.e., $\arg \max_k f_\theta(x_i)_k = \arg \max_k f_\theta(\tilde{x}_i)_k$ for all $\|\delta\|_\infty \leq \epsilon$.

Accordingly, the learning objective is to minimise the worst-case expected loss over $\mathcal{D}$ under all admissible perturbations: $\min_\theta \mathbb{E}_{(x,y) \sim \mathcal{D}} [\max_{\|\delta\|_\infty \leq \epsilon} \ell(f_\theta(x + \delta), y)]$, where ℓ is a classification loss (e.g., cross-entropy). The following sections describe how WAET approximates this minimax objective by generating adversarial examples using gradient-based attacks (e.g., FGSM, BIM, and PGD) and by adaptively weighting harder adversarial samples during training to improve robustness without sacrificing clean performance.

Weighted Adversarial Example Training: Weighted Adversarial Example Training (WAET) hardens a classifier against multiple gradient-based attacks while keeping the inference pipeline unchanged. As illustrated in Fig. 1, the training loop consists of two branches that share the backbone f_θ . The clean branch processes each unperturbed input

Algorithm 1: Weighted Adversarial Example Training (WAET)

Input: Training set $\mathcal{D}$, model f_θ , attack routine $\mathcal{A}$, budget ϵ , mixing coefficient λ , temperature β

Output: Optimised parameters θ

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1 foreach mini-batch  $\mathcal{B} = \{(x_i, y_i)\}_{i=1}^B$  do
2   Generate  $\tilde{x}_i \leftarrow \mathcal{A}(x_i, y_i; f_\theta, \epsilon)$  for all  $i$ 
3   Compute  $L_i^{\text{adv}} \leftarrow \ell(f_\theta(\tilde{x}_i), y_i)$  and set  $\hat{w}_i \leftarrow \exp(\beta L_i^{\text{adv}})$  for all  $i$ 
4   Normalise  $w_i \leftarrow \hat{w}_i / \sum_{j=1}^B \hat{w}_j$  for all  $i$ 
5   Evaluate  $L_{\text{clean}} \leftarrow \frac{1}{B} \sum_{i=1}^B \ell(f_\theta(x_i), y_i)$  and
       $L_{\text{adv}} \leftarrow \sum_{i=1}^B w_i \ell(f_\theta(\tilde{x}_i), y_i)$ 
6   Update  $\theta \leftarrow \theta - \eta \nabla_\theta ((1 - \lambda) L_{\text{clean}} + \lambda L_{\text{adv}})$ 

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(x_i, y_i) and computes the standard classification loss, thereby preserving accuracy on benign data. In parallel, the adversarial branch perturbs x_i using a K -step projected gradient descent (PGD) procedure constrained by an ℓ_∞ budget ϵ to produce an adversarial example $\tilde{x}_i$. The resulting adversarial samples are passed through the same backbone, and their losses are used to improve robustness.

To integrate information from both branches, WAET minimises a weighted combination of clean and adversarial losses. Let B denote the batch size. We define $\mathcal{L}(\theta) = (1 - \lambda) L_{\text{clean}}(\theta) + \lambda L_{\text{adv}}(\theta)$, with mixing coefficient $\lambda \in [0, 1]$ and $L_{\text{adv}}(\theta) = \sum_{i=1}^B w_i \ell(f_\theta(\tilde{x}_i), y_i)$. Unlike standard adversarial training, WAET assigns a sample-specific weight w_i to each adversarial loss, reflecting the difficulty of that example for the current model. These weights are obtained by normalising exponentiated adversarial losses: $\hat{w}_i = \exp(\beta \ell(f_\theta(\tilde{x}_i), y_i))$, $w_i = \frac{\hat{w}_i}{\sum_{j=1}^B \hat{w}_j}$, where the temperature $\beta > 0$ controls the sharpness of the distribution. When β is small, the weights are close to uniform; larger values concentrate weight on high-loss (hard) adversarial examples, steering optimisation toward the most vulnerable regions of the decision boundary.

Fig. 1 visualises this workflow. The PGD generator in the adversarial branch iteratively (i) computes the gradient of the loss in terms of the input, (ii) takes one step in the ascent direction, and (iii) projects back into the ϵ -ball, yielding an adversarial example $\tilde{x}_i$. Clean and adversarial losses are computed on their respective branches, scaled by $(1 - \lambda)$ and λ , summed, and then used to update θ via backpropagation. Because both branches share parameters, the resulting model balances clean accuracy and robustness, and no additional computation is required at deployment.

The overall training procedure is summarised in Algorithm 1. For each mini-batch, we generate adversarial examples, compute per-sample adversarial losses and weights, evaluate the resulting weighted objective, and update θ . By blending clean and adversarial objectives and emphasising harder adversarial examples, WAET improves robustness against FGSM, BIM, and PGD while avoiding the high inner-loop cost typically associated with multi-step PGD training.

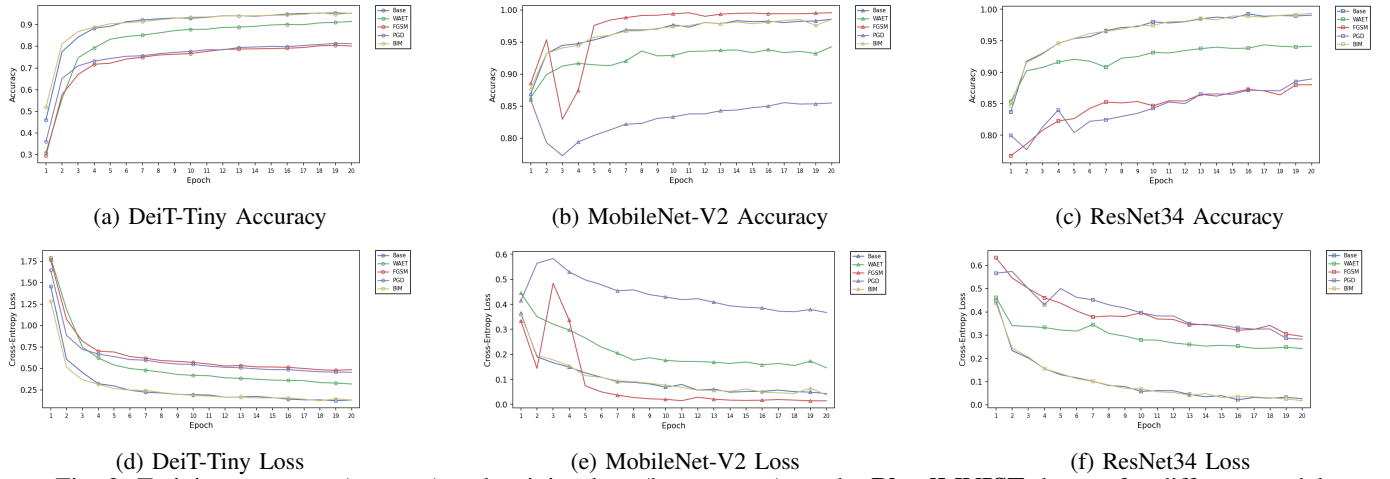


Fig. 2: Training accuracy (top row) and training loss (bottom row) on the **BloodMNIST** dataset for different model architectures, with perturbation coefficient $\epsilon = 0.03$.

TABLE I: Complete results on **BloodMNIST** (mean $\pm$ std over 3 runs). Clean accuracy is reported once, while robust accuracy is evaluated under FGSM, BIM, and PGD with perturbation budgets $\epsilon \in \{0.01, 0.03, 0.05\}$.

Model	Strategy	Clean	$\epsilon = 0.01$			$\epsilon = 0.03$			$\epsilon = 0.05$		
			FGSM	BIM	PGD	FGSM	BIM	PGD	FGSM	BIM	PGD
MobileNet-V2 (2.23M)	Base	0.9600 $\pm$ 0.0000	0.2719 $\pm$ 0.0379	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.1867 $\pm$ 0.0551	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.1729 $\pm$ 0.0085	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
	WAET	0.8987$\pm$0.0194	0.8902$\pm$0.0473	0.5828$\pm$0.4876	0.5828$\pm$0.4876	0.8042$\pm$0.0199	0.5458$\pm$0.4119	0.5458$\pm$0.4119	0.6926$\pm$0.0384	0.4581$\pm$0.3744	0.4581$\pm$0.3744
	FGSM	0.9490 $\pm$ 0.0073	0.7181 $\pm$ 0.0307	0.1618 $\pm$ 0.1229	0.1618 $\pm$ 0.1229	0.9848 $\pm$ 0.0050	0.0022 $\pm$ 0.0025	0.0022 $\pm$ 0.0025	0.8879 $\pm$ 0.0759	0.0002 $\pm$ 0.0003	0.0002 $\pm$ 0.0003
	PGD	0.8879 $\pm$ 0.0089	0.8613 $\pm$ 0.0080	0.8607 $\pm$ 0.0082	0.8607 $\pm$ 0.0082	0.8034 $\pm$ 0.0101	0.7943 $\pm$ 0.0130	0.7943 $\pm$ 0.0130	0.7289 $\pm$ 0.0148	0.6944 $\pm$ 0.0149	0.6944 $\pm$ 0.0149
	BIM	0.9515 $\pm$ 0.0102	0.2700 $\pm$ 0.0665	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.1859 $\pm$ 0.0465	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.1472 $\pm$ 0.0465	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
DeiT-Tiny (5.53M)	Base	0.9363 $\pm$ 0.0068	0.6420 $\pm$ 0.0149	0.5591 $\pm$ 0.0031	0.5591 $\pm$ 0.0031	0.3878 $\pm$ 0.0605	0.1692 $\pm$ 0.0171	0.1692 $\pm$ 0.0171	0.3020 $\pm$ 0.0619	0.0649 $\pm$ 0.0390	0.0649 $\pm$ 0.0390
	WAET	0.9141$\pm$0.0100	0.8807$\pm$0.0086	0.8802$\pm$0.0088	0.8802$\pm$0.0088	0.8013$\pm$0.0081	0.7918$\pm$0.0088	0.7918$\pm$0.0088	0.7062$\pm$0.0112	0.6614$\pm$0.0096	0.6614$\pm$0.0096
	FGSM	0.9221 $\pm$ 0.0072	0.8866 $\pm$ 0.0102	0.8858 $\pm$ 0.0107	0.8858 $\pm$ 0.0107	0.7948 $\pm$ 0.0094	0.7757 $\pm$ 0.0123	0.7757 $\pm$ 0.0123	0.6914 $\pm$ 0.0188	0.6174 $\pm$ 0.0134	0.6174 $\pm$ 0.0134
	PGD	0.9010 $\pm$ 0.0062	0.8748 $\pm$ 0.0072	0.8742 $\pm$ 0.0077	0.8742 $\pm$ 0.0077	0.8100 $\pm$ 0.0063	0.8014 $\pm$ 0.0050	0.8014 $\pm$ 0.0050	0.7363 $\pm$ 0.0059	0.7041 $\pm$ 0.0025	0.7041 $\pm$ 0.0025
	BIM	0.9388 $\pm$ 0.0063	0.6407 $\pm$ 0.0130	0.5555 $\pm$ 0.0031	0.5555 $\pm$ 0.0031	0.4044 $\pm$ 0.0713	0.1924 $\pm$ 0.0533	0.1924 $\pm$ 0.0533	0.3208 $\pm$ 0.0745	0.0731 $\pm$ 0.0460	0.0731 $\pm$ 0.0460
ResNet-34 (21.29M)	Base	0.9393 $\pm$ 0.0209	0.5038 $\pm$ 0.0115	0.3502 $\pm$ 0.0231	0.3502 $\pm$ 0.0231	0.1344 $\pm$ 0.0382	0.0079 $\pm$ 0.0076	0.0079 $\pm$ 0.0076	0.0592 $\pm$ 0.0323	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
	WAET	0.8982$\pm$0.0098	0.8714$\pm$0.0132	0.8705$\pm$0.0136	0.8705$\pm$0.0136	0.8028$\pm$0.0143	0.7926$\pm$0.0138	0.7926$\pm$0.0138	0.7101$\pm$0.0187	0.6729$\pm$0.0180	0.6729$\pm$0.0180
	FGSM	0.9279 $\pm$ 0.0160	0.8836 $\pm$ 0.0152	0.8810 $\pm$ 0.0135	0.8810 $\pm$ 0.0135	0.7491 $\pm$ 0.0200	0.7158 $\pm$ 0.0182	0.7158 $\pm$ 0.0182	0.6053 $\pm$ 0.0295	0.5266 $\pm$ 0.0261	0.5266 $\pm$ 0.0261
	PGD	0.9030 $\pm$ 0.0136	0.8799 $\pm$ 0.0109	0.8791 $\pm$ 0.0105	0.8791 $\pm$ 0.0105	0.8154 $\pm$ 0.0037	0.8072 $\pm$ 0.0016	0.8072 $\pm$ 0.0016	0.7360 $\pm$ 0.0095	0.7053 $\pm$ 0.0007	0.7053 $\pm$ 0.0007
	BIM	0.9458 $\pm$ 0.0060	0.5545 $\pm$ 0.0113	0.4106 $\pm$ 0.0220	0.4106 $\pm$ 0.0220	0.1924 $\pm$ 0.0073	0.0095 $\pm$ 0.0099	0.0095 $\pm$ 0.0099	0.1044 $\pm$ 0.0043	0.0004 $\pm$ 0.0007	0.0004 $\pm$ 0.0007

IV. PERFORMANCE EVALUATION

We evaluate WAET on BloodMNIST from the MedMNIST benchmark suite [4]. BloodMNIST contains microscopic blood cell images spanning eight categories. The dataset is separated into training, validation, and test sets. Also, images are resized to 224×224 pixels and normalised prior to training.

Experiments are conducted on three architectures representative of edge-friendly and server-scale deployments: *DeiT-Tiny* [5], *MobileNet-V2* [6], and *ResNet-34* [7]. WAET is compared against four training strategies: (i) *Base*, empirical risk minimisation on clean data; (ii) *FGSM* training [8]; (iii) *BIM* training [9] with three iterations; and (iv) *PGD* training [10] with ten iterations, which serves as a strong adversarial-training baseline. Unless stated otherwise, WAET uses a mixing coefficient $\lambda = 0.5$ and a weight temperature $\beta = 5$.

We test each model against three first-order attacks (FGSM, BIM, and PGD) under ℓ_∞ budgets $\epsilon \in \{0.01, 0.03, 0.05\}$. During training, we use a three-step PGD to generate adversarial examples for WAET and PGD baselines (three steps for BIM, one for FGSM). We choose three-step PGD to balance robustness and efficiency: adding more PGD iterations yields

negligible robustness gains but substantially increases training cost [11]. During evaluation, we adopt a stronger ten-step PGD. All models are trained for 20 epochs with SGD and a cosine schedule for learning rate.

We evaluate the trained models on several complementary metrics: (i) *Clean Accuracy*: Classification accuracy on unperturbed test images, reflecting baseline predictive performance. (ii) *Robust Accuracy*: Classification accuracy on testing images are perturbed by a specific attack and budget (FGSM, BIM, or PGD), quantifying resilience to gradient-based adversaries. (iii) *Mean and Standard Deviation*: Each experiment is repeated 3 times with random seeds and the mean accuracy and its standard deviation across runs are reported to capture variability. (iv) *Training Dynamics*: We track training accuracy and loss curves over epochs to verify that optimisation remains stable. (v) *Cross-attack Generalisation*: We compare robust accuracies across FGSM, BIM, and PGD to assess whether robustness to one attack transfers to others or whether training overfits to a single attack type. (vi) *Sensitivity to Perturbation Budgets*: We evaluate robust accuracy at multiple ϵ values (0.01, 0.03, 0.05) to understand how performance degrades as the adversary’s strength increases. (vii) *Statistical Significance*: We perform two-sample *t*-tests to assess whether WAET’s im-

provements over the baseline methods are statistically significant. Following conventional practice, results with $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$ correspond to significance at the 5%, 1% and 0.1% levels, respectively; the exact p-values are reported in the significance heatmaps.

All models are trained for 20 epochs using SGD with a cosine learning rate schedule. WAET and PGD training use a three-step PGD (three steps for BIM and one step for FGSM) to reduce training cost, while evaluation uses a ten-step PGD. The same perturbation budget ϵ is applied during training and evaluation, and other hyperparameters (batch size and learning rate) are kept consistent across baselines unless otherwise noted.

Results and Analysis: We assess whether WAET improves robustness against multiple gradient-based attacks without compromising clean accuracy. Fig. 2 shows smooth training dynamics for $\epsilon = 0.03$: training accuracy increases steadily and loss decreases monotonically, indicating that emphasising harder adversarial examples does not destabilise optimisation. WAET trains for the same number of epochs as the baselines and does not introduce additional inner-loop iterations. Single-step FGSM training suffers from catastrophic overfitting: the model overfits to fixed-magnitude perturbations, so FGSM accuracy spikes even as robustness against multi-step PGD collapses. Table I summarise clean and robust accuracies across models, attacks, and budgets. WAET consistently improves robust accuracy across FGSM, BIM, and PGD while keeping clean accuracy within a few percentage points of the Base model. These results indicate that WAET improves robustness without sacrificing clean performance.

Second, we evaluate cross-attack generalisation. Models trained against a single attack often generalise poorly: for instance, FGSM training yields high FGSM robustness but low BIM and PGD robustness, and BIM or PGD training may overfit to the attack budget. By generating adversarial examples via PGD and reweighting harder samples, WAET achieves consistently high robust accuracy across FGSM, BIM, and PGD, as shown in Table I. This suggests that the decision boundary obtained by WAET is less sensitive to the specific attack used at test time.

Third, we examine sensitivity to the perturbation budget. As expected, robust accuracy decreases as ϵ increases. WAET degrades more gracefully than the baselines; for example, Table I show that ResNet-34 on BloodMNIST retains 67.3% PGD robustness at $\epsilon = 0.05$, whereas standard training collapses to zero. These observations indicate that WAET yields decision boundaries that are less sensitive to perturbation magnitude. We note that some WAET configurations exhibit large standard deviations, indicating unstable training or insufficient runs.

In summary, the experimental evaluation shows that WAET effectively addresses all three of our research questions. It improves robustness across multiple gradient-based attacks (FGSM, BIM, and PGD) without compromising clean accuracy, exhibits stable and efficient training dynamics, and in most cases yields statistically significant gains over standard adversarial-training baselines. These results suggest that

WAET offers a practical defence for IoMT models under first-order ℓ_∞ attacks.

V. FINAL REMARKS

In this paper, we tackled adversarial robustness for IoMT medical-imaging pipelines, where diagnostic data traverse untrusted networks and devices operate under strict resource constraints. We introduced Weighted Adversarial Example Training (WAET), a training-only defence that combines clean and adversarial losses and upweights the hardest samples, so inference remains as fast as a standard classifier. Evaluations on BloodMNIST across DeiT-Tiny, MobileNet-V2 and ResNet-34 show that WAET consistently enhances robustness against FGSM, BIM and PGD attacks at multiple budgets while largely preserving clean accuracy; in some cases it matches or surpasses full multi-step PGD training. Learning curves indicate stable optimisation, and significance tests across three runs suggest the gains are not due to chance. Overall, WAET offers a practical lightweight defence for IoMT models, though our evaluation is limited to first-order attacks and small sample sizes.

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